SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Constituent of Symbiosis International (Deemed University), Pune

Name of Student: Sumit kr. Mahato **PRN: 25070122195**

Subject: Programming with Java **Sem: III** **AY: 2026-27**

Assignment No: 3

TITLE: Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

Theory

Method overloading allows multiple methods to have the same name with different parameter lists, enabling compile-time polymorphism. The static keyword is used for variables and methods that belong to the class instead of individual objects. Static members share common data among all objects and reduce memory usage. These concepts improve code flexibility, readability, and efficient resource management.

Example code

Calculator.java

```java
class Calculator {

    // Static variable
    static int count = 0;

    // Constructor
    Calculator() {
        count++;
    }

    // Overloaded methods
    int add(int a, int b) {
        return a + b;
    }

    double add(double a, double b) {
        return a + b;
    }

    int add(int a, int b, int c) {
        return a + b + c;
    }

    // Static method
    static void displayCount() {
        System.out.println("Objects Created: " + count);
    }

    public static void main(String[] args) {

        Calculator c1 = new Calculator();
        Calculator c2 = new Calculator();

        System.out.println("Addition of two integers: " + c1.add(25, 30));
        System.out.println("Addition of two doubles: " + c1.add(20.5, 30.5));
        System.out.println("Addition of three integers: " + c1.add(20, 30, 40));

        Calculator.displayCount();
    }
}
```

Output

```
java — -zsh — 80×24
sumit@Sumit java % javac Calculator.java
sumit@Sumit java % java Calculator
Addition of two integers: 55
Addition of two doubles: 51.0
Addition of three integers: 90
Objects Created: 2
sumit@Sumit java %
```

Exercises

1. Develop a **Calculator program** using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

Source Code

Users > sumit > Documents > java > Calculator1.java > Calculator1 > add(int, int)

```java
public class Calculator1 {

    // Static variable
    static int count = 0;

    // Constructor
    Calculator1() {
    }

    // Overloaded methods
    int add(int a, int b) {
        count++;
        return a + b;
    }

    double add(double a, double b) {
        count++;
        return a + b;
    }

    int add(int a, int b, int c) {
        count++;
        return a + b + c;
    }

```

```java
    // Static method
    static void displayCount() {
        System.out.println("Total Calculations: " + count);
    }

    Run | Debug
    public static void main(String[] args) {

        Calculator1 c1 = new Calculator1();
        Calculator1 c2 = new Calculator1();

        System.out.println("Addition of two integers: " + c1.add(a: 25, b: 30));
        System.out.println("Addition of two doubles: " + c1.add(a: 20.5, b: 30.5));
        System.out.println("Addition of three integers: " + c1.add(a: 20, b: 30, c: 40));

        Calculator1.displayCount();
    }
}
```

Output

```
java — -zsh — 102×46

sumit@Sumit java % javac Calculator1.java
sumit@Sumit java % java Calculator1
Addition of two integers: 55
Addition of two doubles: 51.0
Addition of three integers: 90
Total Calculations: 3
sumit@Sumit java %
```

2.Develop a **Restaurant Billing Application** where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

Users > sumit > Documents > java > Restaurant.java > Restaurant > Restaurant()

```java
class Restaurant {

    // Static variable
    static int count = 0;

    // Constructor
    Restaurant() {
    }

    // Overloaded methods
    double bill(double amount) {
        count++;
        return amount;                 // Dine-in
    }

    double bill(double amount, double packingCharge) {
        count++;
        return amount + packingCharge; // Takeaway
    }

    double bill(double amount, double packingCharge, double deliveryCharge) {
        count++;
        return amount + packingCharge + deliveryCharge; // Delivery
    }

    // Static method
    static void displayCount() {
        System.out.println("Total Orders: " + count);
    }

```

```java
    Run | Debug
    public static void main(String[] args) {

        Restaurant r1 = new Restaurant();
        Restaurant r2 = new Restaurant();

        System.out.println("Dine-in Bill: " + r1.bill(amount: 500));
        System.out.println("Takeaway Bill: " + r1.bill(amount: 500, packingCharge: 30));
        System.out.println("Delivery Bill: " + r1.bill(amount: 500, packingCharge: 30, deliveryCharge: 50));

        Restaurant.displayCount();
    }
}
```

Output:


java — -zsh — 102×46
sumit@Sumit java % javac Restaurant.java
sumit@Sumit java % java Restaurant
Dine-in Bill: 500.0
Takeaway Bill: 530.0
Delivery Bill: 580.0
Total Orders: 3
sumit@Sumit java %